input



File: power_input.kicad_sch

buckboost



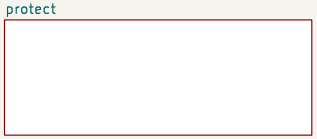
File: buckboost.kicad_sch

mcu



File: mcu.kicad_sch

protect



File: protect_sense.kicad_sch

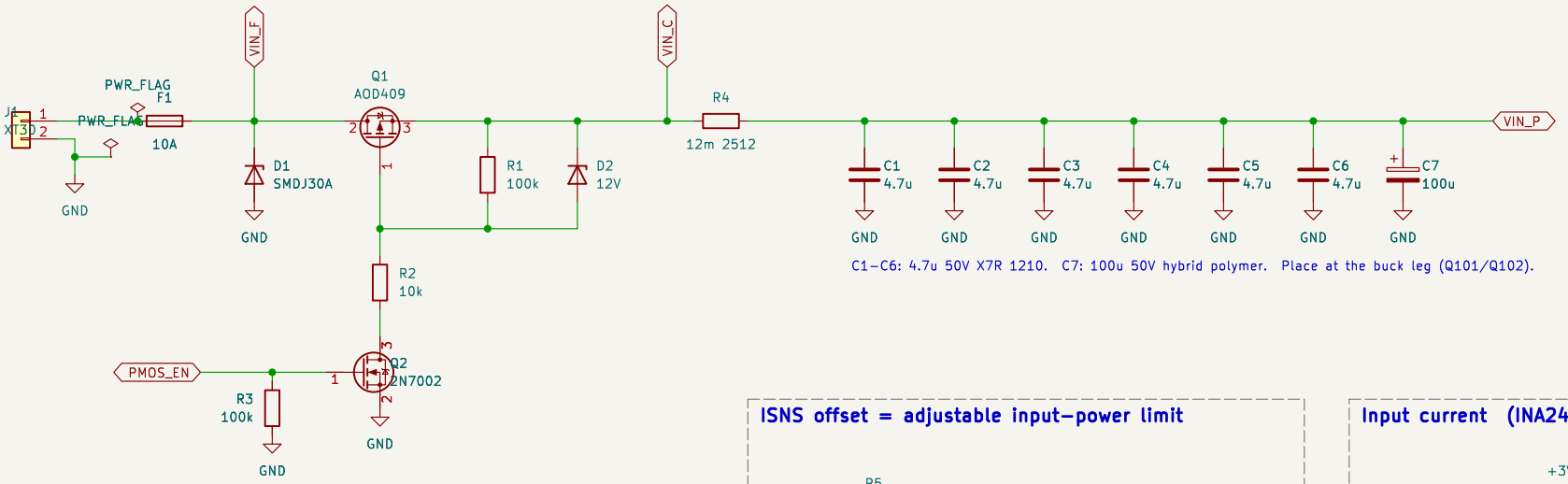
Sheet: /
File: control_board.kicad_sch

Title: Rev B control board (root)

Size: A4	Date:	**Rev:**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/5

1/4 Power input (PM chassis output 22–26 V)

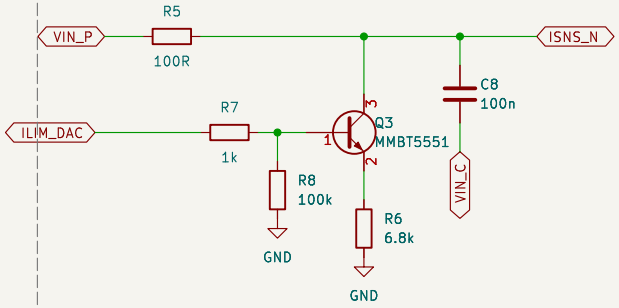
Q1 blocks back-feed from converter/bank into the PM output (body diode conducts input -> board only). Q1 stays OFF unless PMOS_EN is high.
R4 = LM5176 input shunt (ISNS limit 50 mV -> 4.2 A). Kelvin-route ISNS+ (VIN_C), ISNS− (via R5) and the INA240 inputs from the R4 pads.



C1–C6: 4.7u 50V X7R 1210. C7: 100u 50V hybrid polymer. Place at the buck leg (Q101/Q102).

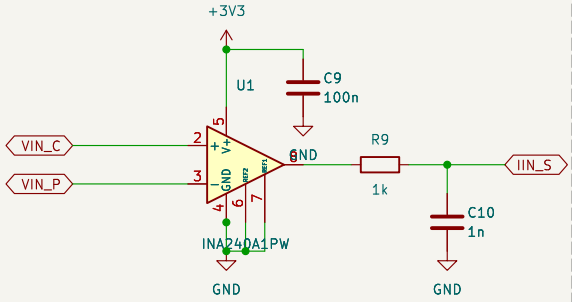
PMOS_EN (sheet 4) = MCU EN_PMOS AND no input OV. R3 keeps Q1 off at power-up.

ISNS offset = adjustable input-power limit



I_ofs = (V_DAC − 0.65 V)/R6, max ~0.4 mA -> +40 mV on ISNS
=> limit 4.2 A ... ~0.8 A. MCU closes the loop with IIN_S.

Input current (INA240A1, G=20 -> 0.24 V/A)

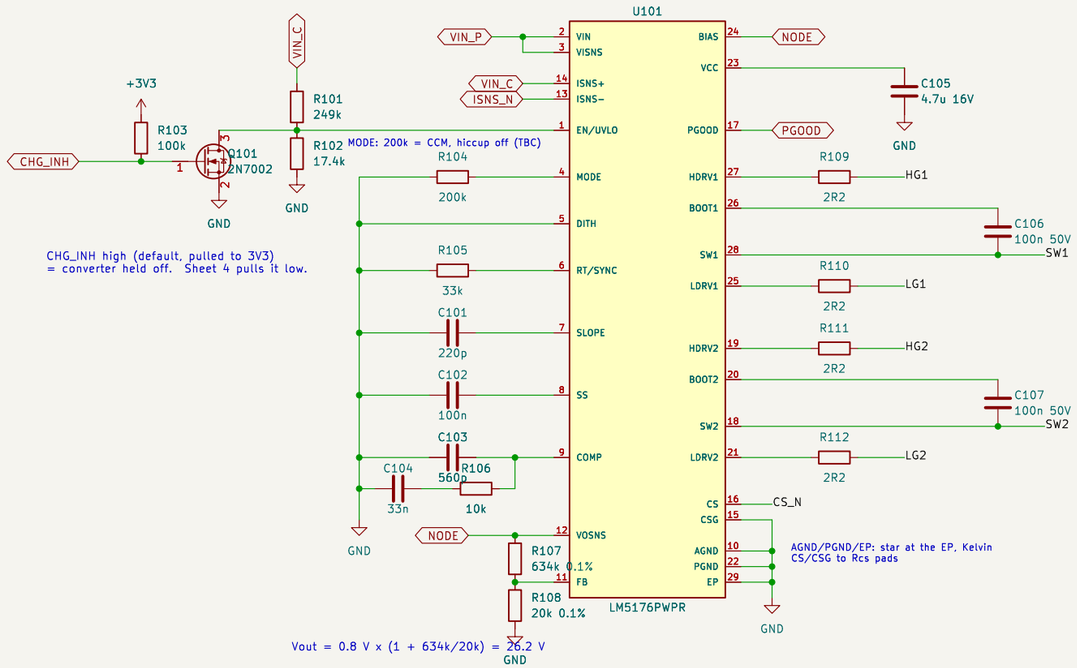


NAGOYA SHACHIHOKO	
Sheet: /input/	
File: power_input.kicad_sch	
Title: Rev B control board – 1/4 power input	
Size: A4	Date:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 2/5

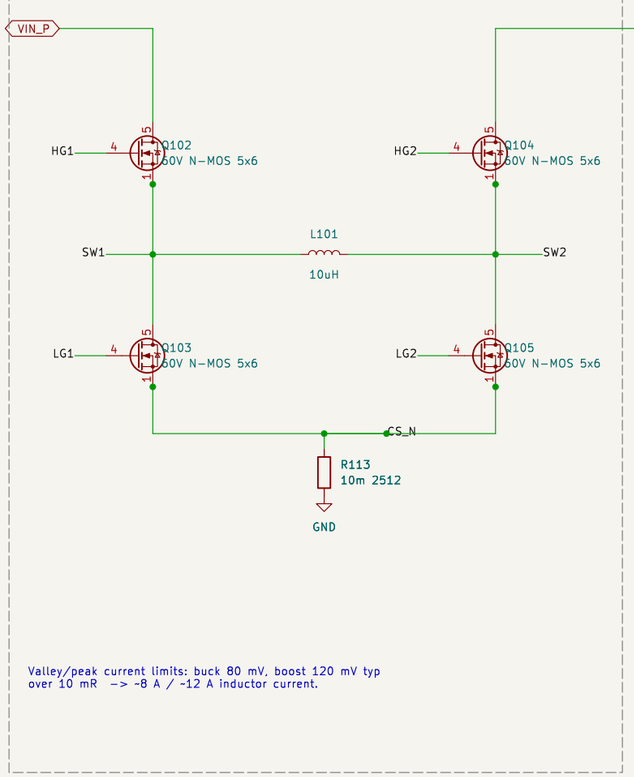
Rev: B-prov

2/4 LM5176 4-switch buck-boost charger (VIN_P 22-26 V -> NODE 0-26.2 V, ~100 W)

NODE = chassis output and capacitor bank (via CM01). fsw ~248 kHz (RT 33k), Vout 26.2 V (634k/20k), UVLO ~19 V, CCM, hiccup off (MODE 200k: confirm in datasheet).
Power FETs: 60 V class (input TVS clamps up to ~48 V). Gate nets HG1/LG1/HG2/LG2 and SW1/SW2 are local labels; keep gate loops short in layout.

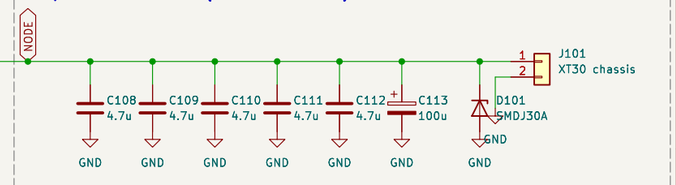


Power stage

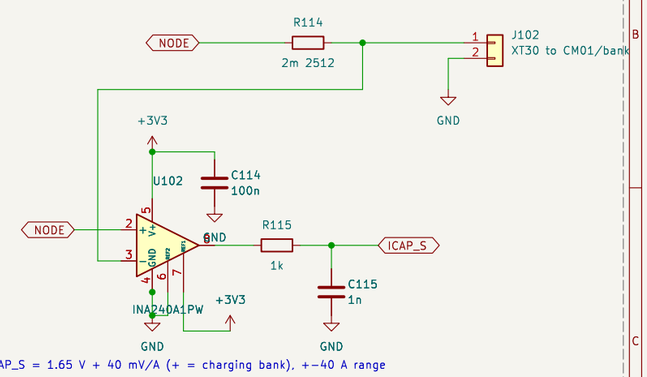


Valley/peak current limits: buck 80 mV, boost 120 mV typ
over 10 mR -> ~8 A / ~12 A inductor current.

Output node NODE (chassis + bank)



C: 4.7u 50V X7R 1210 at the boost leg.
D: clamps regen spikes if the bank is cut off.



ICAP_S = 1.65 V + 40 mV/A (+ = charging bank), +-40 A range

NAGOYA SHACHIHOKO

Sheet: /buckboost/
File: buckboost.kicad_sch

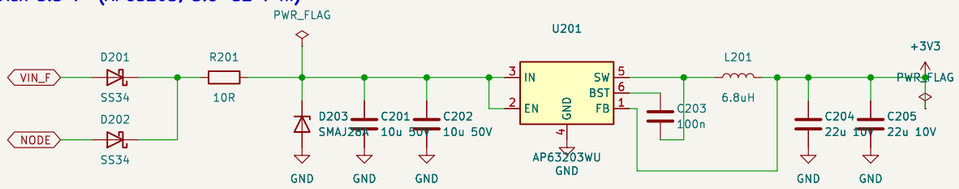
Title: Rev B control board - 2/4 buck-boost charger

Size: A3	Date:	Rev: B-prov
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/5

3/4 Aux 3.3 V supply, MCU (STM32G431CBT6), CAN, referee UART

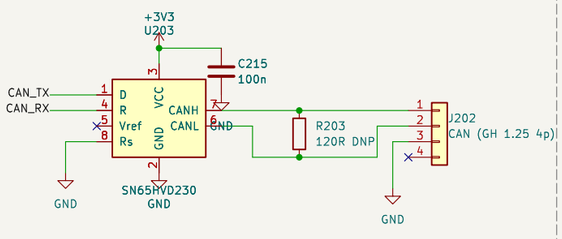
3V3 is fed from PM input OR the bank (NODE), so the MCU stays alive while the robot is unpowered (bank drain ~0.15 W). MCU runs only the slow outer loop.

Aux 3.3 V (AP63203, 3.8–32 V in)

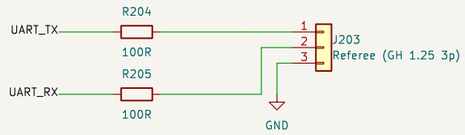


D: SS34 (40V 3A). R 10R limits SMAJ28A current so AUX_IN stays below the AP63203 35 V abs max.

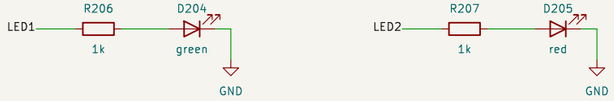
Robot CAN (SN65HVD230, 1 Mbps)



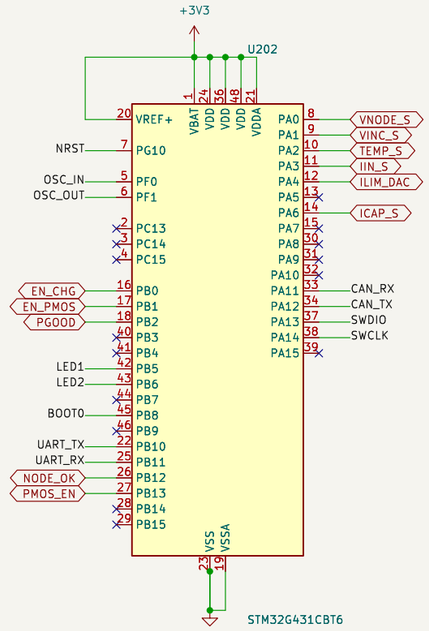
Referee UART (3.3 V TTL: confirm)



Status LEDs

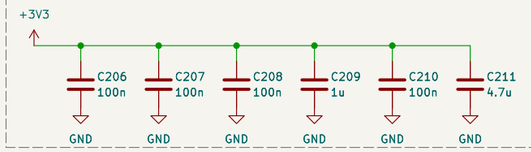


MCU

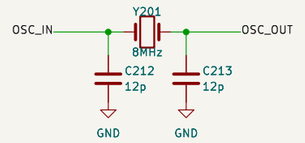


PA4 = DAC1_OUT1 (ILIM_DAC), PA6 = ADC2_IN3, PA11/12 = FDCAN1, PB10/11 = USART3.

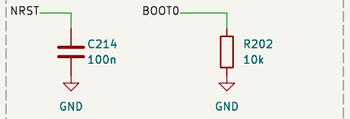
MCU decoupling (one 100n per VDD pin, 1u at VDDA)



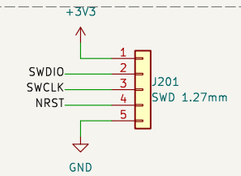
8 MHz crystal (FDCAN clock)



Reset / BOOT0



SWD

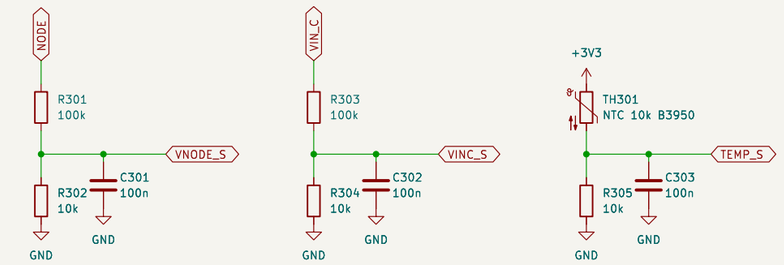


NAGOYA SHACHIHOKO	
Sheet: /mcu/	
File: mcu.kicad_sch	
Title: Rev B control board — 3/4 MCU and aux power	
Size: A3	Date:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 4/5

4/4 Sensing, hardware over-voltage protection, converter enable logic

The converter runs only if PMOS_EN (MCU EN_PMOS and no input OV) AND NODE_OK (no node OV) AND EN_CHG (MCU). Everything defaults to OFF. The comparators work without the MCU.

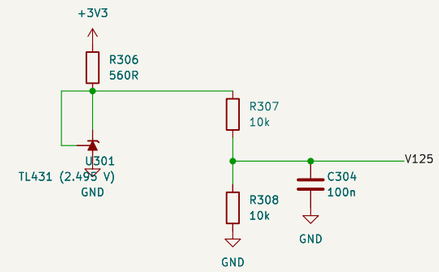
Voltage / temperature sensing (ADC)



27 V -> 2.45 V

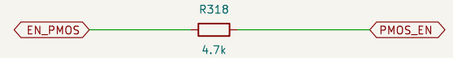
26 V -> 2.36 V

1.25 V comparator reference

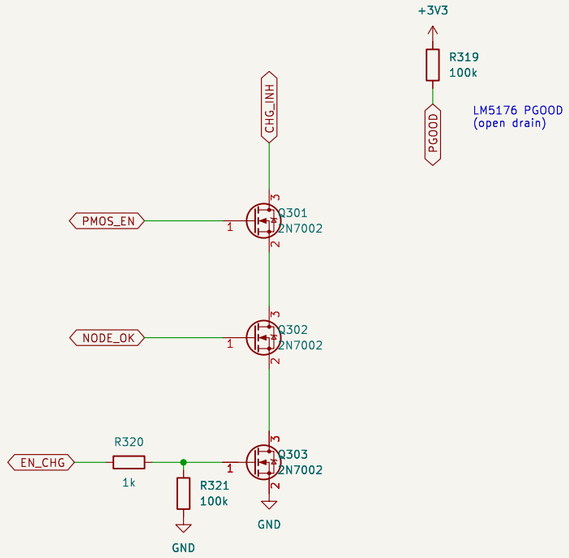


1.2475 V: below the LM393 input range limit (Vcc − 1.5 V).

Converter enable logic (default OFF)

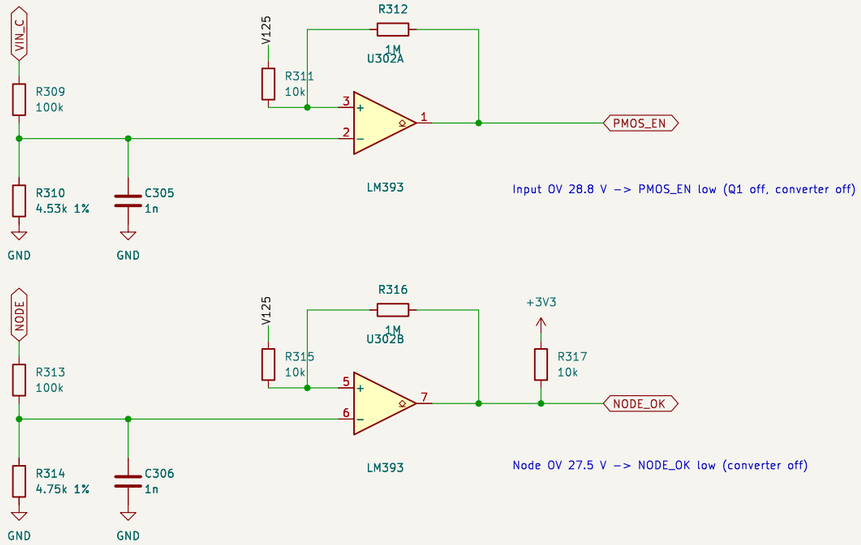


MCU sources PMOS_EN through 4.7k; comparator A sinks it on input OV.
100k pull-down at Q2 (sheet 1) keeps it low at reset.



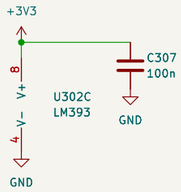
Qa.Qb.Qc all ON -> CHG_INH pulled low -> LM5176 EN released.
Any of them OFF -> CHG_INH high (100k to 3V3, sheet 2) -> converter off.

Hardware over-voltage comparators (LM393, open collector)



Input OV 28.8 V -> PMOS_EN low (Q1 off, converter off)

Node OV 27.5 V -> NODE_OK low (converter off)



Hysteresis: 10k/1M on the + input, about 0.7 V at the monitored rail. 1n filters on the dividers.

NAGOYA SHACHIHOKO		
Sheet: /protect/		
File: protect_sense.kicad_sch		
Title: Rev B control board — 4/4 sensing and protection		
Size: A3	Date:	**Rev: B-prov**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/5